

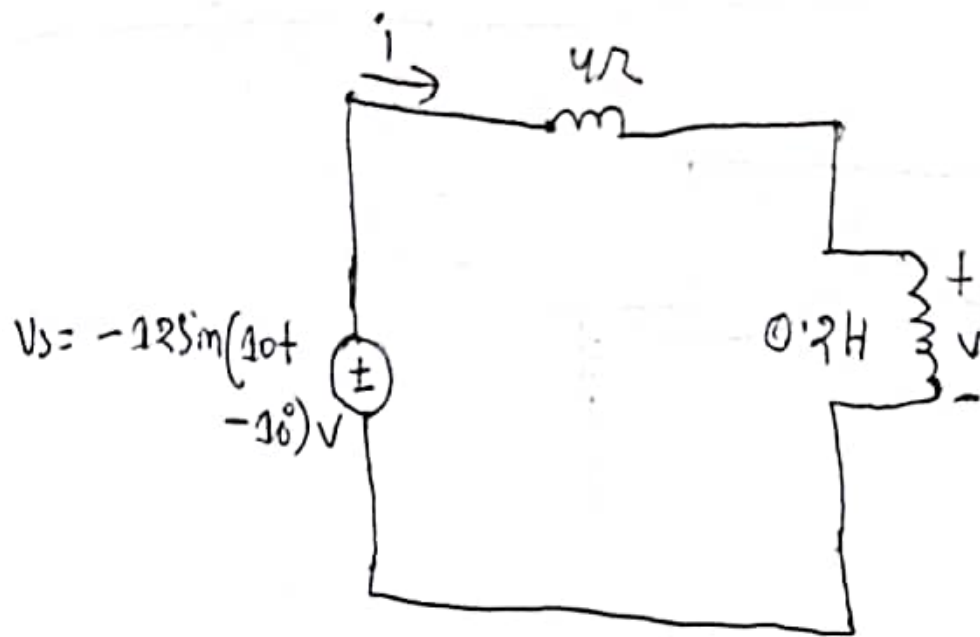
Oli Ahmed

2019-3-60-073

CSE-809, Section-3

Final Exam

Ans to the ques no-2

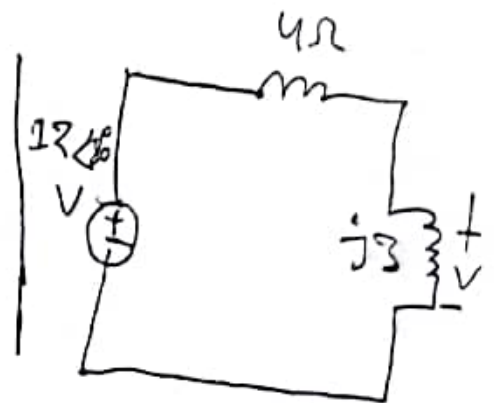


$0.2 \text{ H} :$

$$Z_L = j\omega L$$

$$= j10 \times 0.2$$

$$= j2 \Omega$$



$$V_s = -12 \sin(10t - 10^\circ)$$

$$= 12 \cos(10t - 10^\circ + 90^\circ)$$

$$= 12 \cos(10t + 180^\circ)$$

$$V = 12 \angle 80^\circ \text{ V}$$

$$\begin{aligned} Z &= Z_R + Z_L \\ &= (4 + j2) \Omega \end{aligned}$$

$$\begin{aligned} I &= \frac{V_s}{Z} = \frac{12 \angle 80^\circ}{4 + j2} \\ &= 2.683 \angle 53.434^\circ \end{aligned}$$

$$\therefore i(t) = 2.683 \cos(10t + 53.434^\circ) \text{ A}$$

Now,

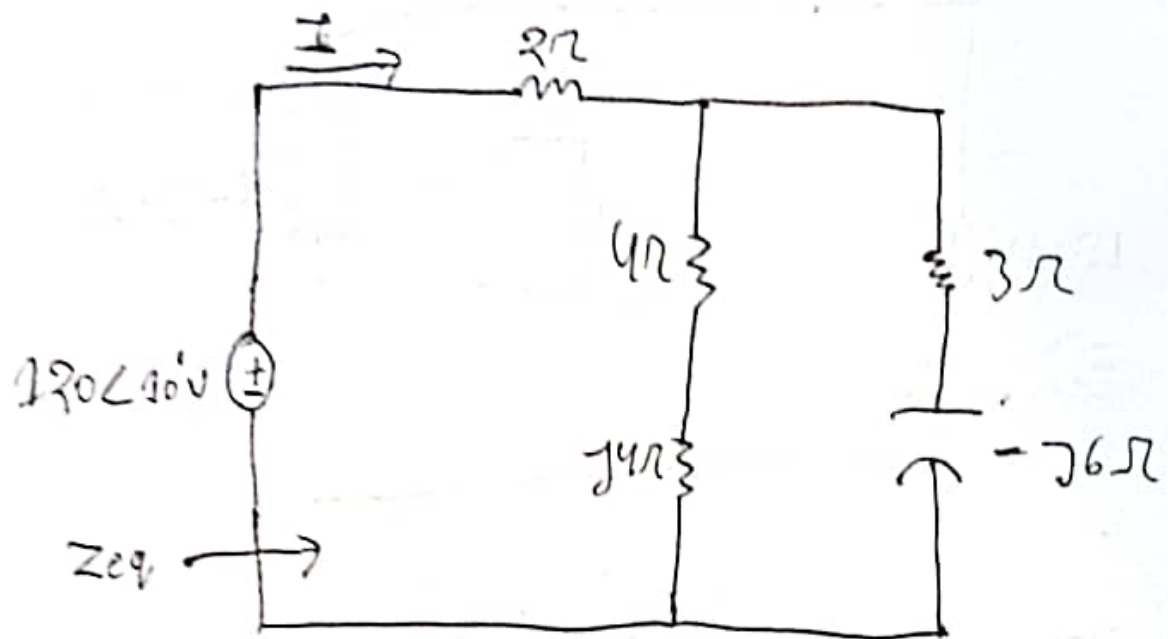
$$\begin{aligned} V &= I Z_L \\ &= (2.683 \angle 53.434^\circ) \times j2 \\ &= 5.366 \angle 143.434^\circ \text{ V} \end{aligned}$$

$$\therefore V(t) = 5.366 \cos(20t + 143.434^\circ) \text{ V}$$

$$\therefore W(t) = 0.366 \cos(20t + 143.434^\circ) \text{ V}$$

As-

Ans to the question no-2



Hence,

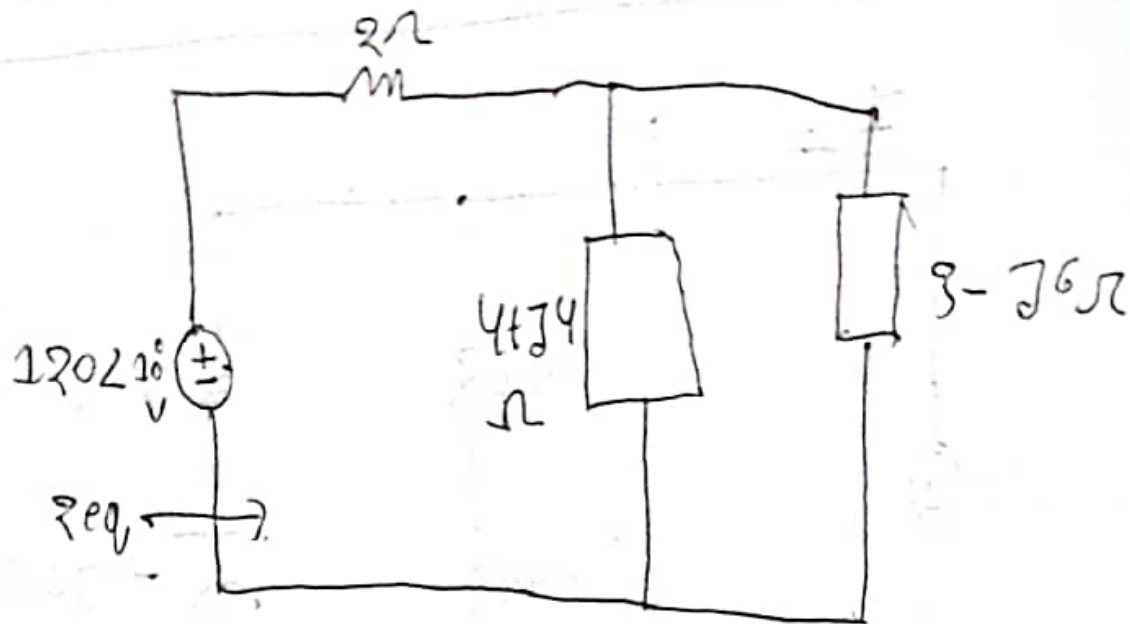
3Ω and $-j6 \Omega$ are in series

$$\text{---} = 3 - j6$$

4Ω and $j4 \Omega$ are in series.

$$= 4 + j4$$

Now,



$$Z_{eq} = \frac{(4 + j4) \times (3 - j6)}{(4 + j4) + (3 - j6)} + 2$$

$$= \frac{(4 + j4)(3 - j6)}{7 - j2} + 2$$

$$= 7.208 - j0.226 \Omega$$

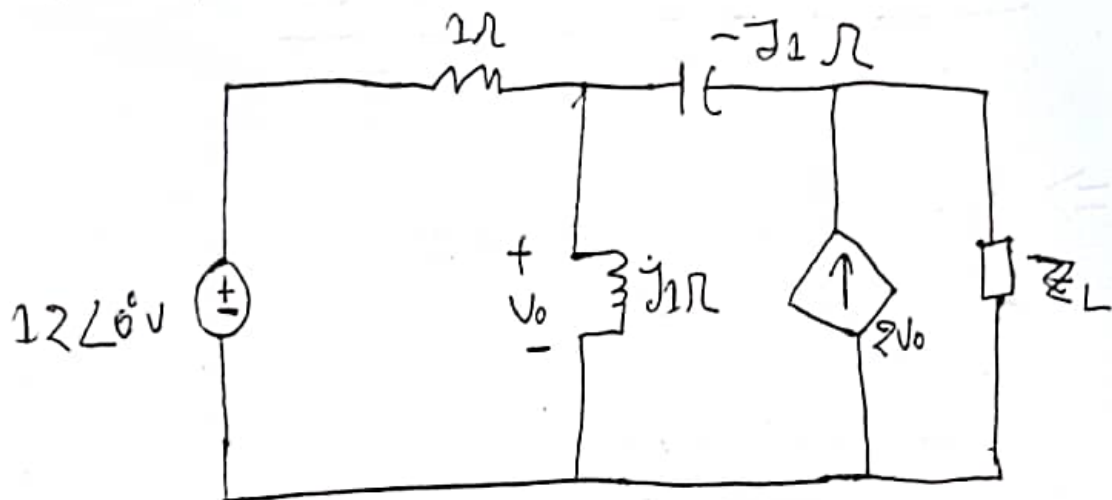
$$\therefore I = \frac{V}{Z_{eq}} = \frac{120 \angle 10^\circ}{7.208 - j0.226}$$

$$= 16.64 \angle 11.776^\circ \text{ A}$$

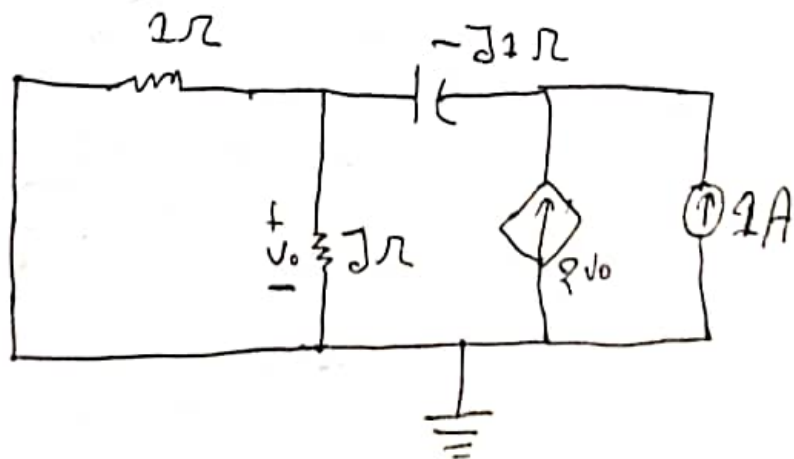
(Ans.)

Ans to the ques no-5

Hence given,



for Z_{eq} :



KCL at node-1:

$$\frac{V_{o1}}{1} + \frac{V_{o1}}{j} = \frac{V_2 - V_1}{j}$$

$$\Rightarrow V_1 = jV_2 \dots (1)$$

KCL at node-2:

$$1 + 2V_o = \frac{V_2 - V_1}{-j}$$

$$\Rightarrow 1 + 2V_1 = \frac{V_2 - V_1}{-j} \quad [V_o = V_1]$$

$$\Rightarrow 1 = jV_2 - (2+j)V_1 \dots (2)$$

Substituting Eq(1) into (2),

$$1 = jV_2 - (2+j)(j)V_2$$

$$= (1-j)V_2$$

$$\Rightarrow V_2 = \frac{1}{1-j}$$

Now,

$$Z_{eq} = \frac{V_2}{1} = \frac{1+j}{1}$$

$$= 0.5 + j0.5$$

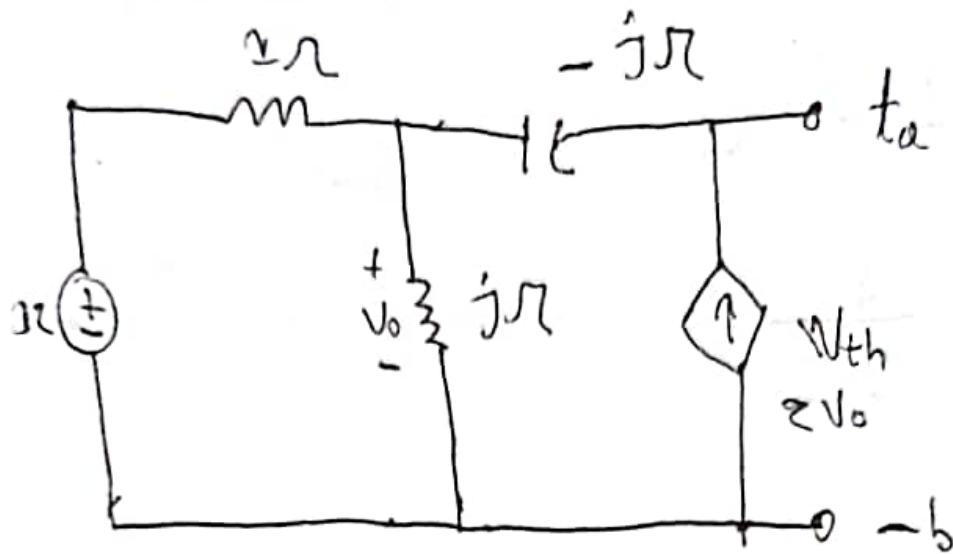
So,

Load resistance,

$$Z_L = Z_{eq}$$

$$= (0.5 - j0.5) \Omega$$

Now, for $V_{th} = 0$



$$-2V_0 + \frac{V_0 - 12}{1} + \frac{V_0}{j} = 0$$

$$\Rightarrow V_0 = \frac{-12}{1+j}$$

$$\Rightarrow -V_0 - (-j \times 2V_0) + V_{th} = 0$$

Now,

$$V_{th} = (1 - j2)V_0$$

$$= \frac{(-12)(1 - j2)}{1+j}$$

Now,

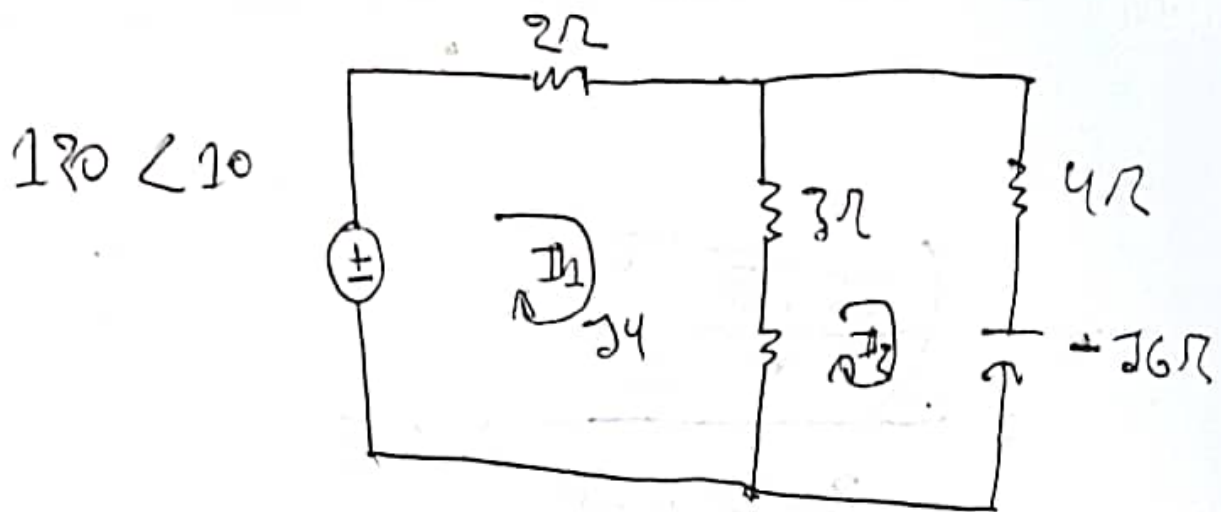
$$P_{\text{max}} = \frac{\left| \left[\frac{V_{th}}{0.5 + j0.5 + 0.5 - j0.5} \right] \right|^2}{2} \times 0.5$$

$$= \frac{\left(\frac{12\sqrt{5}}{\sqrt{2}} \right)^2}{2 \times (2 \times 0.5)^2} \times 0.5$$

$$\text{---} = 90 \text{ W}$$

Ans.

Ans to the ques no-9



KVL at mesh 1:

$$2I_1 + 3(I_1 - I_2) + j4(I_1 - I_2) = 120\angle 10^\circ$$

$$\Rightarrow 2I_1 + 3I_1 - 3I_2 + j4I_1 - j4I_2 = 120\angle 10^\circ$$

$$\Rightarrow I_1(2+3+j4) + I_2(-3-j4) = 120\angle 10^\circ$$

$$\Rightarrow I_1(5+j4) + I_2(-3-j4) = 120\angle 10^\circ$$

KVL at mesh 2:

$$3(I_2 - I_1) + j4(I_2 - I_1) + 4I_2 - j6I_2 = 0$$

$$\Rightarrow 3I_2 - 3I_1 + j4I_2 - j4I_1 + 4I_2 - j6I_2 = 0$$

$$\Rightarrow I_1(-3 - j4) + I_2(3 + j4 + 4 - j6) = 0$$

$$\Rightarrow I_1(-3 - j4) + I_2(7 - j2) = 0 \quad \text{--- (1)}$$

Now,

$$A = \begin{vmatrix} 5 + j4 & -3 - j4 \\ -3 - j4 & 7 - j2 \end{vmatrix}$$

$$= (5 + j4) \times (7 - j2) - (-3 - j4) \times (-3 - j4)$$

$$= 50.358 \angle -6.843$$

$$A_1 = \begin{vmatrix} 120 \angle 10 & -3 - j4 \\ 0 & 7 - j2 \end{vmatrix}$$

$$= 120 \angle 10 \times (7 - j2)$$

$$= 873.613 \angle -5.945$$

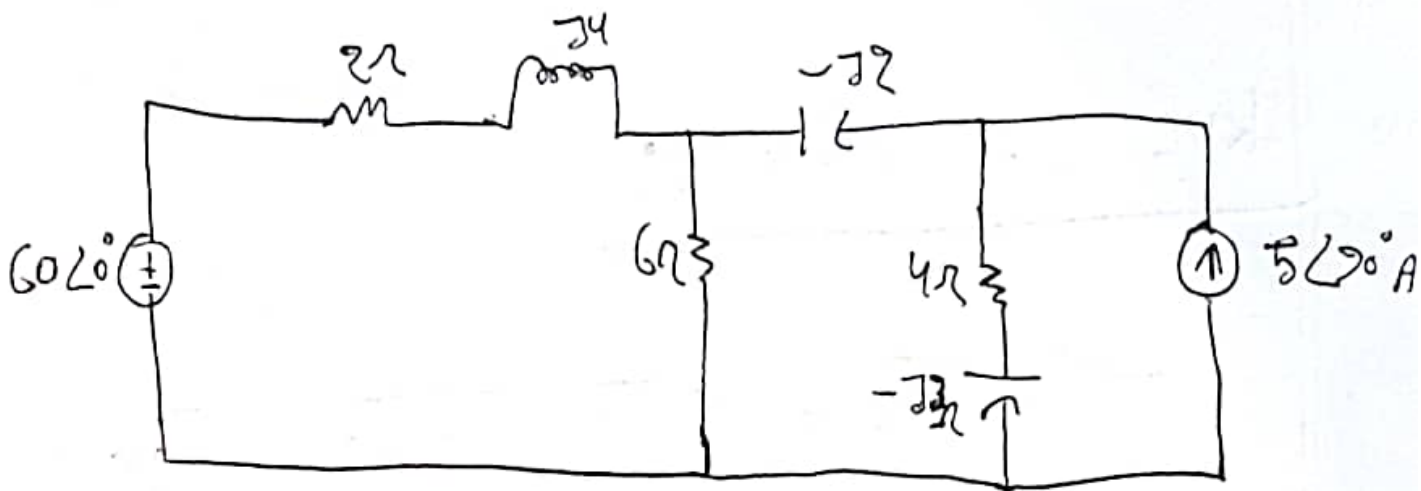
$$\therefore I_1 = \frac{A_1}{\Delta}$$

$$= \frac{873.613 \angle -5.945}{50.358 \angle -6.848}$$

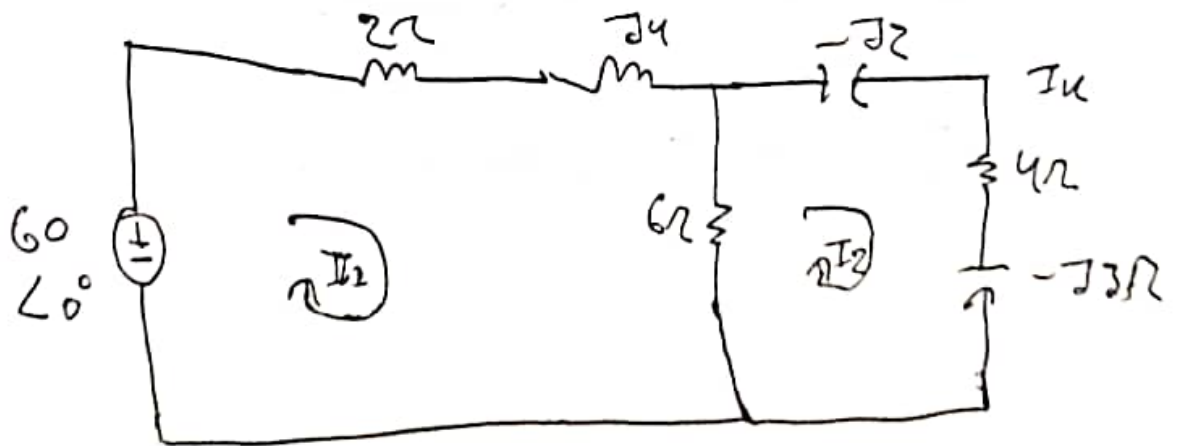
$$= 17.348 \angle 0.898$$

A-

Ans to the question no-4



60∠0° Active:



KVL at mesh 1:

$$2I_1 + j4I_1 + 6I_1 - 6I_2 = 60\angle 0^\circ$$

$$\Rightarrow (8+j4)I_1 - 6I_2 = 60 \angle 0^\circ \text{ --- (I)}$$

KVL at mesh 2:

$$-j2 \cdot I_2 + 4I_2 - j3I_2 + 6I_2 = 0$$

$$\Rightarrow I_2(-j2 + 4 - j3 + 6) - 6I_1 = 0$$

$$\Rightarrow I_2(-j5 + 10) - 6I_1 = 0$$

$$\Rightarrow -I_1 6 + I_2(-j5 + 10) = 0 \text{ --- (II)}$$

Now,

$$A = \begin{vmatrix} 8+j4 & -6 \\ -6 & -j5+10 \end{vmatrix}$$

$$\begin{aligned}
 &= (8+24)(-25+20) - 36 \\
 &= 200 - 36 \\
 &= 64
 \end{aligned}$$

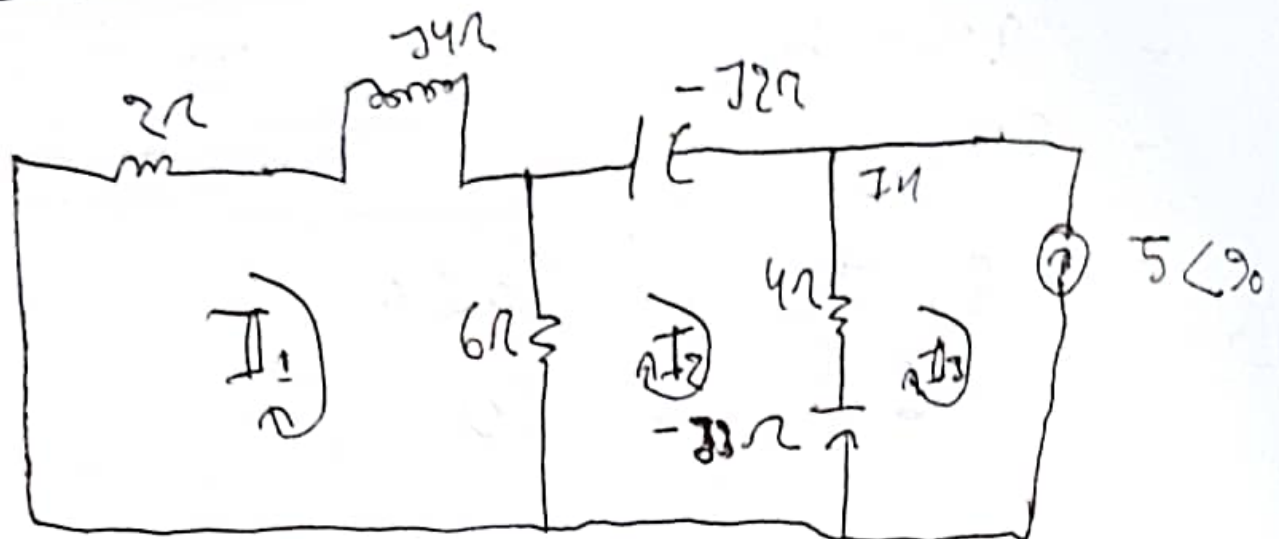
$$\begin{aligned}
 \Delta_2 &= \begin{vmatrix} 8+24 & 60 < 0 \\ -6 & 0 \end{vmatrix} \\
 &= 360
 \end{aligned}$$

Now,

$$\begin{aligned}
 I_2 &= \frac{\Delta_2}{\Delta} = \frac{360}{64} \\
 &= 5.625 \text{ A}
 \end{aligned}$$

$$\therefore I_2 = I_{x1} = 5.625 \text{ A}$$

$5\angle 90$ Active



$$I_{n2} = -I_3$$

$$= 5\angle 90$$

$$\therefore I_n = I_{n1} + I_{n2}$$

$$= 5.625 + 5\angle 90$$

$$= 7.526 \angle 41.633$$

✓